

L2 - 14.1. Exploring Wazuh

Task 1 - Introduction

Wazuh has evolved from a basic intrusion detection system into an all-in-one platform for unified security monitoring and compliance management. A comprehensive, unified security platform that extends beyond EDR to integrate log analysis, vulnerability management, cloud monitoring, and automated compliance tracking.

- Auditing devices for common vulnerabilities
- Monitoring for suspicious activity on the endpoints
- Visualizing complex events into neat dashboards and graphs
- Providing compliance reporting for frameworks like GDPR, HIPAA, NIST

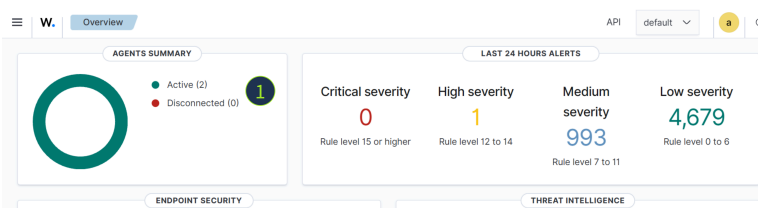
Task 2 - Launch Wazuh Server

Wazuh uses a centralized manager-agent architecture where lightweight agents on endpoints send data to a central server for processing, indexing, and visualization.

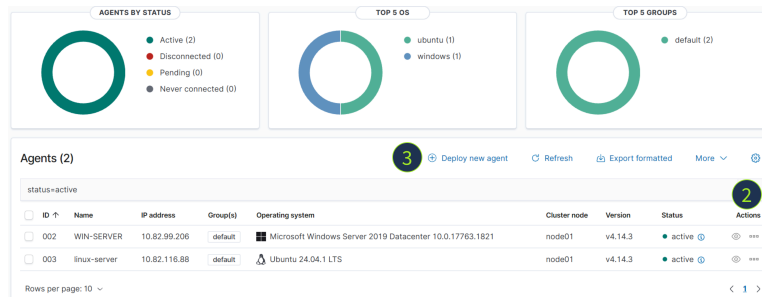
In more technical terms, the single-VM deployment packs the entire Wazuh stack, combining an OpenSearch-based indexer, the custom manager engine, Filebeat data shipping, and an OpenSearch Dashboard interface, into a single host.

Task 3 - Installing Wazuh Agents

Wazuh Agents: Wazuh agents continuously collect endpoint logs and audit system configurations, while filtering by status allows administrators to quickly isolate and troubleshoot disconnected hosts.



You should now see the main Agents menu. Feel free to explore the agents' details by using the Actions menu (2), or see the instructions for deploying new agents (3). Wazuh agents support Windows, Linux, and MacOS, and are fairly easy to install by following the documentation.



Agent Groups: Wazuh agent groups allow administrators to categorize endpoints and automatically apply tailored security configurations, audit policies, and logging rules across specific sets of devices.

The screenshot shows the 'Deploy new agent' form in the Wazuh interface. It includes fields for 'Assign a server address' (10.82.115.80) and 'Remember server address' (checked). Under 'Optional settings', there is a field for 'Assign an agent name' (MY-LAPTOP) and a dropdown for 'Select one or more existing groups' (default). A red box highlights the 'Select one or more existing groups' dropdown. At the bottom, there is a section for 'Run the following commands to download and install the agent:' with a code block containing the necessary commands.

Answer the questions below

What is the status of the agents managed by this Wazuh?

Disconnected

Agents (2)

1 agent selected

status=disconnected

☐	ID ↑	Name	IP address	Group(s)	Operating system
☒	002	WIN-SERVER	10.82.99.206	default	 Microsoft Win
☐	003	linux-server	10.82.116.88	default	 Ubuntu 24.04

Rows per page: 10 ▾

Click on the Windows agent. What is the CPU field value?

AMD EPYC 7571

System inventory

Cores
2

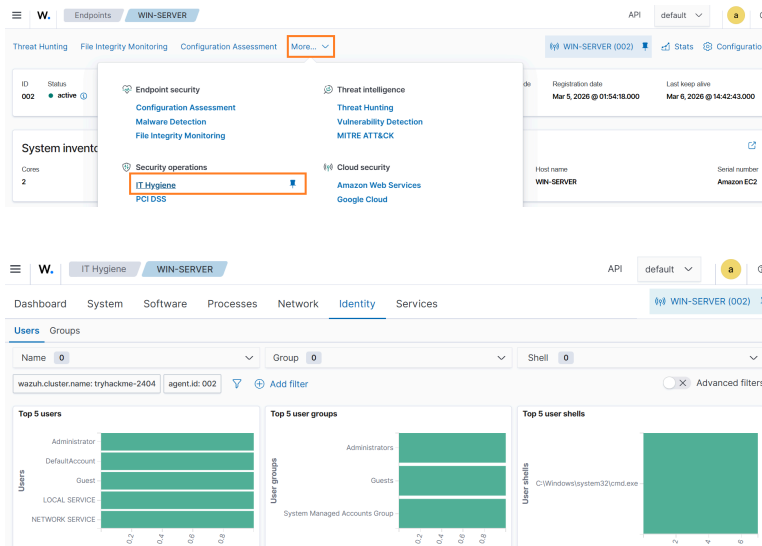
Memory
4GB

CPU
AMD EPYC 7571

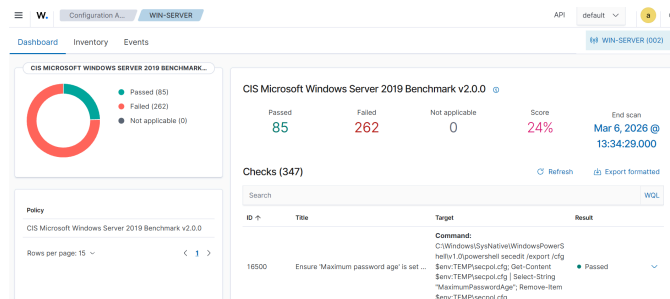
Task 4 - Configuration Assessment

Wazuh and IT Hygiene: Once the agent is installed, Wazuh runs lots of security checks and collects a ton of information about the host. One of the most useful categories for IT is the IT Hygiene checks. Select the **WIN-SERVER** agent, click **More**, and open the **IT Hygiene** section. Once you are there, you will see multiple tabs, such as:

- **System:** Hardware, hostname, network interfaces, and OS version
- **Software:** Installed programs, OS patches, and even browser extensions
- **Network:** Listening ports (e.g., RDP), vital for the Exposure Management
- **Identity:** Local users present on the device, including their privileges



Wazuh and CIS Benchmarks: Wazuh is capable of auditing and monitoring an agent's configuration with CIS benchmarks or even custom checklists. Take a look at the image below. When the Wazuh agent is installed on the Windows machine, it runs 347 CIS security checks (typically PowerShell commands) and verifies the results. You can use it to audit your devices for misconfigurations and prepare them for audit. Select the **WIN-SERVER** agent and open its **Configuration Assessment** panel to get started:



Answer the questions below

Open the Windows agent and navigate to its IT Hygiene tab. What custom text editor is installed there (Software > Packages)?

Notepad++

Top 5 Packages	
Package	Count ↓
Microsoft Windows Server 2019 Datacenter 10.0.17763.1821	1461
Notepad++ (64-bit x64)	9

Now open the Linux agent and its Configuration Assessment tab. What is the CIS Benchmark score of the Linux agent (e.g., 65%)?

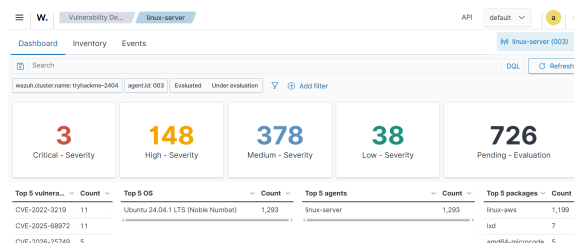
46%

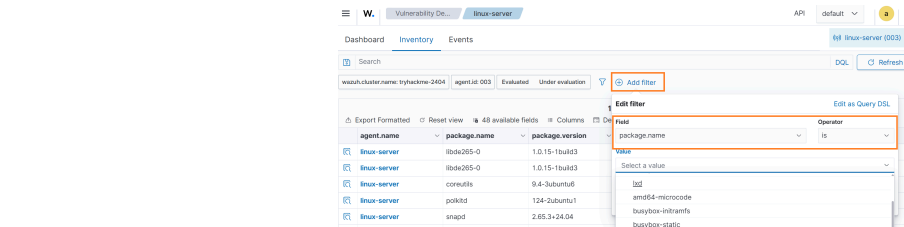
Security Configuration Assessment

Policy	End scan	Pass...	Failed	Not ...	Score
CIS Ubuntu Linux 24.04 LTS Benchmark v1.0.0.	Mar 10, 2026 @ 02:21:26.000	110	127	42	46%

Task 5 - Vulnerability Detection

While Wazuh's Vulnerability Detection module thoroughly indexes all known CVEs for installed software, it flags every technical vulnerability regardless of exploitability, requiring security teams to manually prioritize critical patches based on actual exposure.





Answer the questions below

What is the latest Notepad++ vulnerability found on the Windows agent?

CVE-2026-25926

wazuh.cluster.name: tryhackme-2404	agent.id: 002	Evaluated	Under evaluation	package.name: Notepad++ (64-bit x64)	Add filter
9 hits					
Export Formatted	Reset view	48 available fields	Columns	Density	1 fields sorted
agent.name	package.name	package.version	vulnerability.description	vulnerability.severity	vulnerability.id
WIN-SERVER	Notepad++ (64-bit x64)	8.4	Notepad++ is a free and o...	High	CVE-2026-25926
WIN-SERVER	Notepad++ (64-bit x64)	8.4	Notepad++ versions prior L...	High	CVE-2025-15556

What is the earliest critical vulnerability found on the Linux agent?

CVE-2021-3773

wazuh.cluster.name: tryhackme-2404
agent.id: 003

3
Critical - Severity

Top 5 vulnerabilities	Count
CVE-2021-3773	1
CVE-2024-56180	1
CVE-2025-27558	1

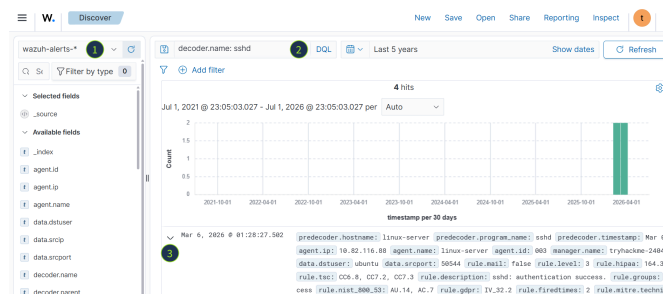
Task 6 - Logging and Reporting

Logging With Wazuh: The core Wazuh capability is to collect and parse security logs from the agents. With some tweaks, Wazuh becomes a powerful SIEM, similar to Elasticsearch and Kibana. Let's start by checking the authentication logs from the **linux-server**. First, open the **top-left menu** > **Explore** dropdown > **Discover** page. This is the panel where all raw logs can be viewed, including the configuration assessment and vulnerability detection results:



Next, let's explore the Discover menu:

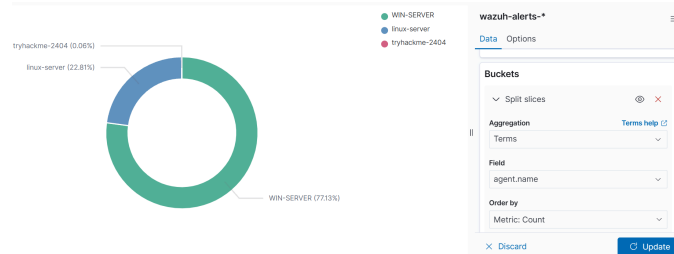
1. Index pattern selector and field search. Useful to learn which fields are present in the events.
2. Search query input and time selector. To view SSH logs, please apply these settings: **Search query:** `decoder.name: sshd` **Time range:** `Last 5 years`
3. The actual events. You can expand each event by clicking the toggle near the timestamp.



Logs vs Rules vs Alerts: Wazuh processes raw events through rule sets, discarding non-matching logs and only generating alerts when an event triggers a specific rule with an assigned severity level from 1 to 15.

Dashboards and Reports: While Wazuh has tons of prebuilt dashboards for routine tasks, you might want to create a custom one at some point. To begin with, you can open the **top-left menu** > **Explore** > **Visualize** > **Create Visualization**.

From there, select **Pie**, choose the **wazuh-alerts-*** source, add the **Split slices** bucket, and aggregate by **agent.name**, as shown in the image below. Then, click **Update**. Don't forget to set the global time range to **Last 5 years**.

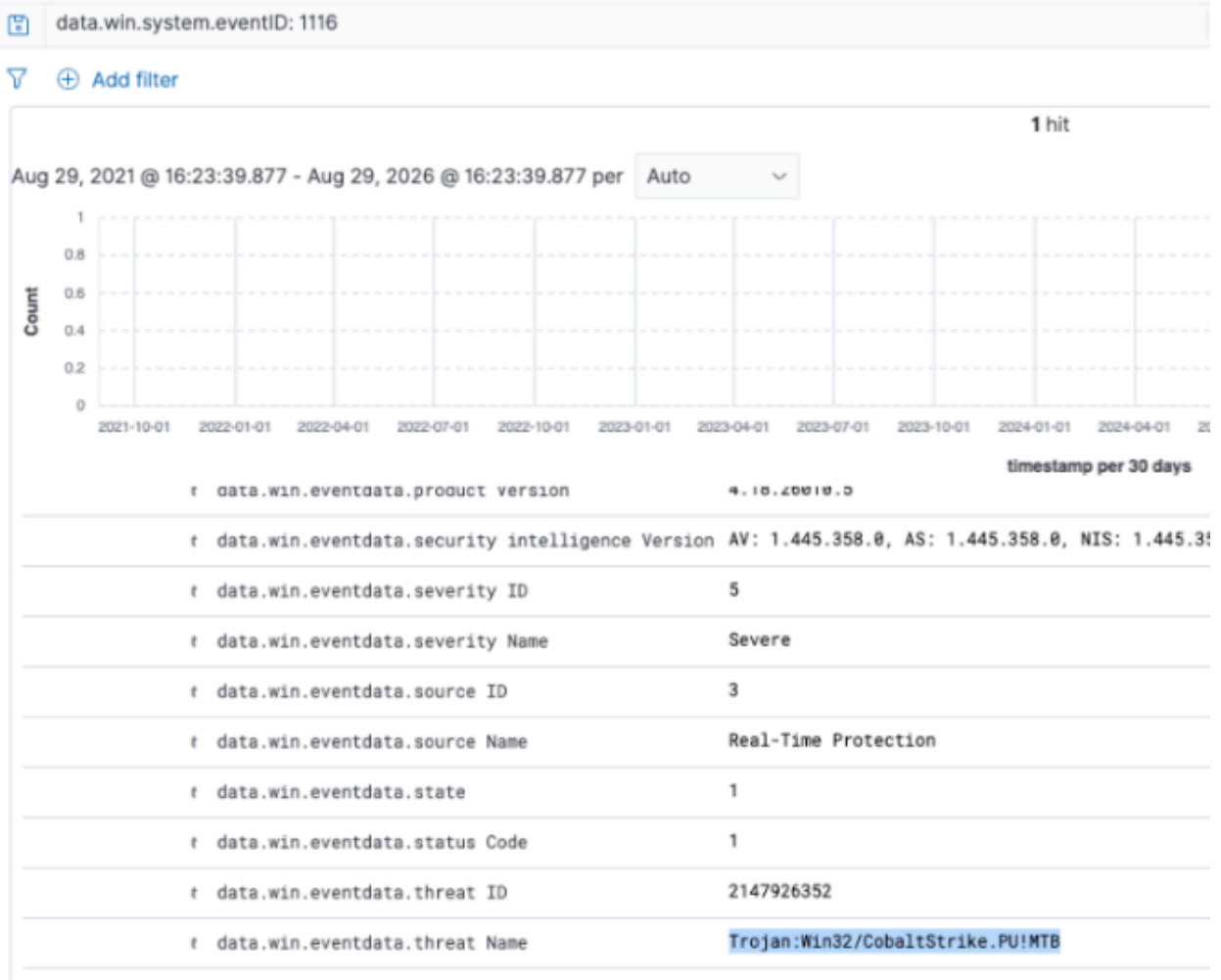


You can combine multiple visualization panels into a single dashboard. You can explore the prepared **Demo Dashboard** by navigating to the **top-left menu** > **Explore** > **Dashboards** menu. The menu might be overwhelming at first, but overall, Wazuh has very powerful reporting capabilities.

Answer the questions below

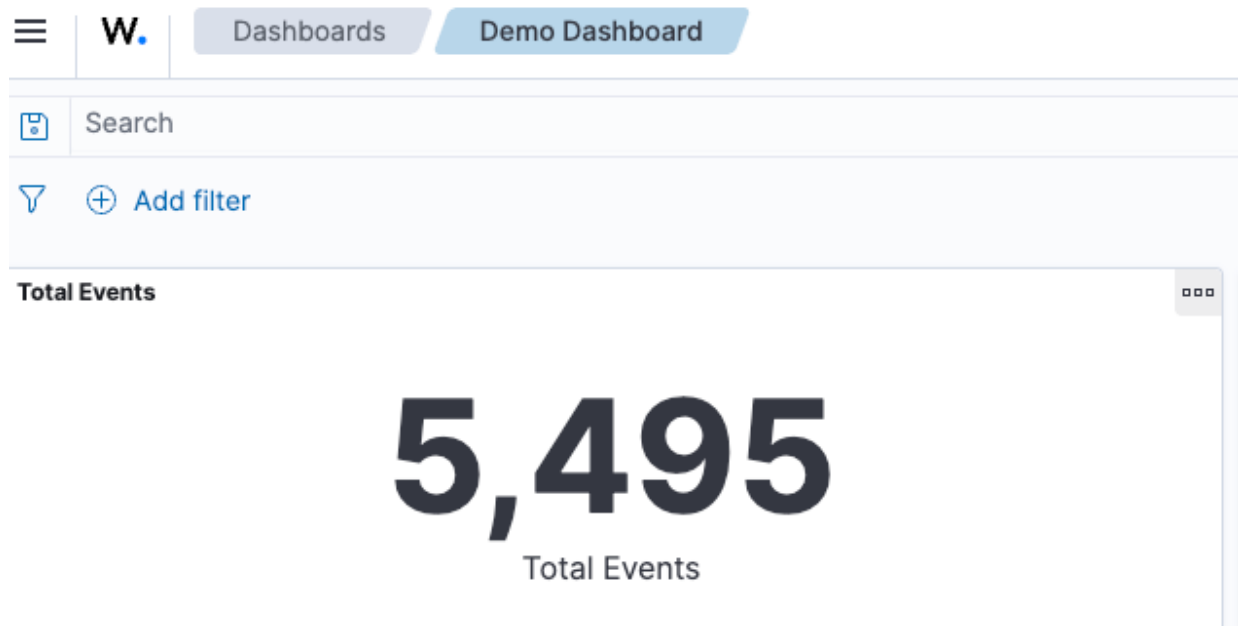
Navigate to Discover and search for Defender alerts for the last 5 years. You can use the query: `data.win.system.eventID: 1116`. What is the threat name of the detected malware?

Trojan:Win32/CobaltStrike.PU!MTB



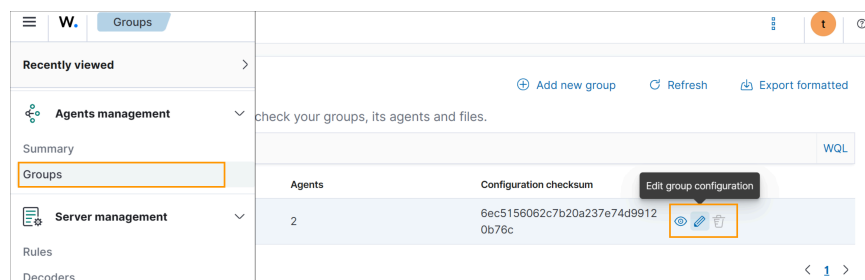
Navigate to the Demo Dashboard. How many total events are shown?

5495



Task 7 - Custom Log Collection

Agent Configuration: Wazuh collects basic system logs by default, but that's usually not enough for mature SOC teams. To collect custom logs, you need to specify the requirements in the **Groups** menu, and they will automatically propagate to all active agents. We have already added two custom settings to the **default** group to collect **/var/log/auth.log** from Linux and MS Defender event logs from Windows agents. Check it out by going to **Agents management > Groups** and clicking the pencil icon:



Wazuh Decoders: The agent configuration defines what to collect from the agents. Next, you need to instruct Wazuh on how to process (parse) the incoming raw data, as every source sends the events in its own unique format. The parsing instructions are called decoders, and there are hundreds of default decoders you can find in **Server management > Decoders** menu. We won't explore the

decoders in this room, but here is an example for Sysmon event code 1 for your reference:

A default Wazuh decoder for Sysmon event code 1

```
<decoder name="Sysmon-EventID#1">
  <parent>windows</parent>
  <type>windows</type>
  <prematch>INFORMATION\(\1\)\.+HashType</prematch>
  <regex>Image: (\.*) \s*CommandLine: \.* \s*User: (\.*) \s
*LogonGuid: \S* \s*LogonId: \S* \s*TerminalSessionId: \S* \s*
IntegrityLevel: \.*HashType: \S* \s*Hash: (\S*) \s*ParentProc
essGuid: \S* \s*ParentProcessID: \S* \s*ParentImage: (\.*) \s
*ParentCommandLine:</regex>
```

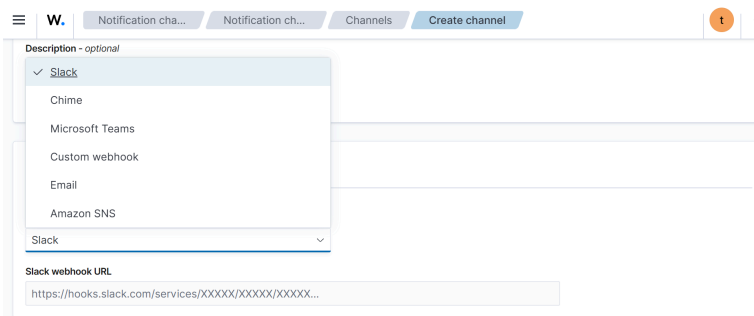
Wazuh Rules: Next, the parsed logs must become alerts to be visible in the Wazuh dashboard. And to create alerts, you need to create Wazuh rules. For example, in order to monitor PowerShell execution on your agents, you will need two rules: the first to alert on all Sysmon event code 1 logs (`level="0"` means informational severity), and the second to alert on any PowerShell execution (`level="12"` means a high-importance event). You can see the existing rules in **Server management > Rules** menu.

Wazuh rules to detect PowerShell execution

```
<group name="sysmon,">
  <rule id="184665" level="0">
    <if_sid>18100</if_sid>
    <match>Microsoft-Windows-Sysmon/Operational: INFORMATION
(1)</match>
    <description>Sysmon - Event Code 1</description>
    <group>sysmon_event1,</group>
  </rule>
```

Alert Notifications: Finally, you can configure notifications if some of your rules are triggered. In the simplest case, you can write a query `rule.level >= 12`, create an alert from it in **Explore > Alerting** menu, and then set up a notification channel

(e.g., Slack message or email). You will receive a notification every time your query matches new events.



Answer the questions below

| How do you call the Wazuh element that extracts fields from raw logs?

Decoder

| What Wazuh alert is more critical, the one with rule.level set to 10 or 15?

Task 8 - Unique Wazuh Features

Explored Features: For IT and GRC, it's:

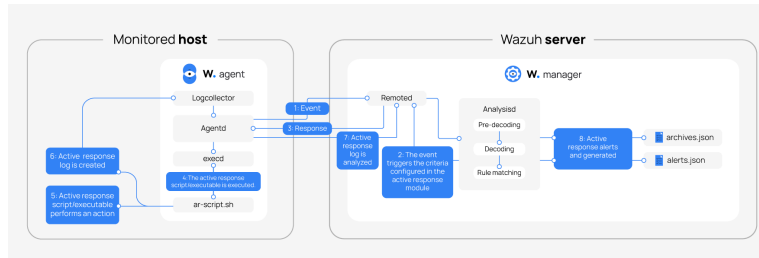
- **Vulnerability Detection** to track software vulnerabilities on the agents
- **Configuration Assessment** to track security misconfigurations on the agents
- **IT Hygiene** to view agent details and list installed apps and browser extensions

There are also a few great features to make Wazuh a SIEM, such as:

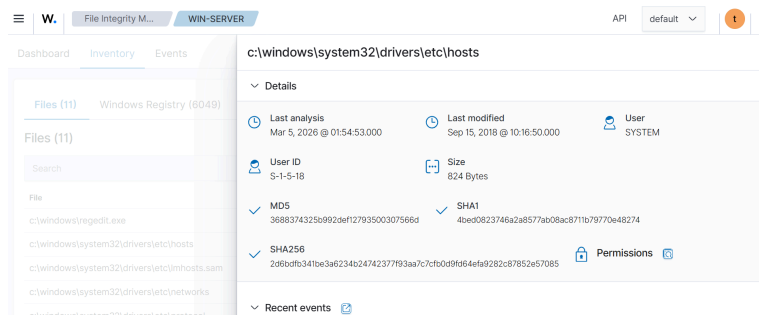
- **Decoders and Rules** to ingest and monitor OS and application logs
- **Discover and Dashboards** to search and visualize the ingested logs

Wazuh Active Response: In response to received events, Wazuh can send a response command (Active Response) to the agent: run a script, delete a file, block a network connection, and more. Just keep in mind that the response is executed with a delay (up to a few minutes), and therefore can't fully replace EDR

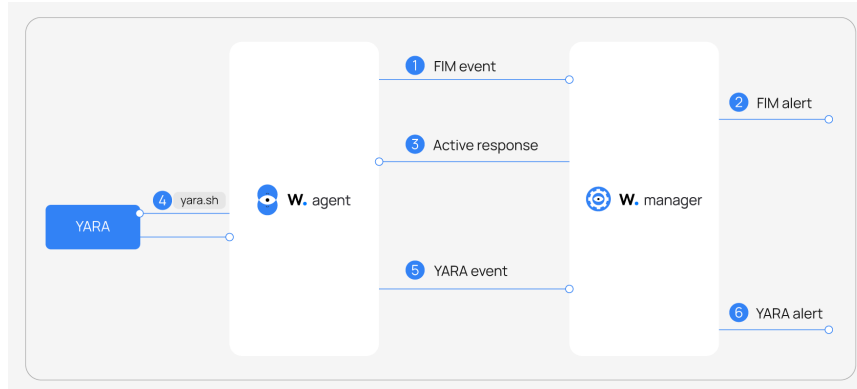
features like real-time process kill or malware quarantine. Still, Active Response can be used as an RMM/MDM for mass software deployment or to support the existing AV/EDR solution.



File Integrity Monitoring: Many government regulations require organizations to monitor critical files for unauthorized changes, including OS configurations, application settings, and VPN profiles. Beyond compliance, companies may also need to enforce strict access controls on sensitive files such as SSH keys. Wazuh addresses both needs through its File Integrity Monitoring (**FIM**) module, which logs every modification to selected files and directories (as well as registry keys on Windows). Explore it yourself under **Endpoint Security > File Integrity Monitoring!**



FIM and Malware Detection: Wazuh goes beyond basic file monitoring with two built-in malware detection features: **YARA** and **VirusTotal** integration. Whenever a monitored file is created or modified, Wazuh automatically scans it using your YARA rules and checks its hash against VirusTotal. A match triggers an alert and, optionally, an Active Response action to contain the threat. Note that you must bring your own YARA rules and VirusTotal key to make the features work.



Wazuh Feature	Description
Cloud Security	Wazuh supports log collection from AWS, Azure, GCP, M365, and GitHub out of the box, and provides nice inventory dashboards for each integration
Compliance Mappings	Every event you receive from Wazuh can be mapped to MITRE ATT&CK, GDPR, PCI DSS, and other compliances. Default Wazuh rules already do this for you.
Osquery Integration	Wazuh can periodically run OSquery queries on the agents and centrally manage OSquery configuration. Extremely useful for custom IT/SOC checks!
Agentless Monitoring	Wazuh can run FIM checks via SSH, without the need for an agent. This is useful for monitoring firewalls (e.g., pfSense) where regular agents can't be installed.

Task 9 - Conclusion

Wazuh is a universal security solution that can fit different needs and be used by different teams. Try installing Wazuh in your home lab, that's pretty fun! Also, if you want to play with Wazuh without ingesting real logs, try ingesting the demo data via **Indexer management > Sample Data**. Just note that some datasets may require an Internet connection to be downloaded.

W.

Sample Data

API default a

Sample security information

Sample data, visualizations and dashboards for security information (File Integrity Monitoring, Amazon Web Services, Office 365, Google Cloud, GitHub, Microsoft Graph API, authorization, ssh, web).

Add data

Sample system inventory

Sample data, visualizations and dashboards for system inventory (groups, hardware, hotfixes, interfaces, networks, packages, ports, processes, protocols, system, users, services, browser extensions).

Add data

Sample Malware Detection

Sample data, visualizations and dashboards for events of Malware Detection (malware, VirusTotal, YARA).

Add data

Sample vulnerability detection inventory

Sample data, visualizations and dashboards for vulnerabilities inventory.

Add data

Sample threat detection and response

Sample data, visualizations and dashboards for threat events of detection and response (Vulnerability Detection, Docker, MITRE ATT&CK).

Add data

L2 - 14.1. Exploring Wazuh

15